



Advanced Linear Algebra: Foundations to Frontiers

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1.3.5 The matrix 2-norm ¶

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Let us instantiate the definition of the vector p norm for the case where $p = 2$, giving us a matrix norm induced by the vector 2-norm or Euclidean norm:

Definition 1.3.5.1. Matrix 2-norm. Define the matrix 2-norm

$\|\cdot\|_2 : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ by

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \max_{\|x\|_2=1} \|Ax\|_2.$$

Remark 1.3.5.2. The problem with the matrix 2-norm is that it is hard to compute. At some point later in this course, you will find out that if A is a Hermitian matrix ($A = A^H$), then $\|A\|_2 = |\lambda_0|$, where λ_0 equals the eigenvalue of A that is largest in magnitude. You may recall from your prior linear algebra experience that computing eigenvalues involves computing the roots of polynomials, and for polynomials of degree three or greater, this is a nontrivial task. We will see that the matrix 2-norm plays an important role in the theory of linear algebra, but less so in practical computation.

Example 1.3.5.3. Show that

$$\left\| \begin{pmatrix} \delta_0 & 0 \\ 0 & \delta_1 \end{pmatrix} \right\|_2 = \max(|\delta_0|, |\delta_1|).$$

▼ Solution

Handouts/Week1/1.3.5TwoNormOf2x2DiagMatrix.pdf]
Handouts/Week1/1.3.5TwoNormOf2x2DiagMatrix.tex

Remark 1.3.5.4. The proof of the last example builds on a general principle: Showing that $\max_{x \in D} f(x) = \alpha$ for some function $f : D \rightarrow \mathbb{R}$ can be broken down into showing that both

$$\max_{x \in D} f(x) \leq \alpha$$

and

$$\max_{x \in D} f(x) \geq \alpha.$$

In turn, showing that $\max_{x \in D} f(x) \geq \alpha$ can often be accomplished by showing that there exists a vector $y \in D$ such that $f(y) = \alpha$ since then

$$\alpha = f(y) \leq \max_{x \in D} f(x).$$

We will use this technique in future proofs involving matrix norms.

Homework 1.3.5.1. Let $D \in \mathbb{C}^{m \times m}$ be a diagonal matrix with diagonal entries $\delta_0, \dots, \delta_{m-1}$. Show that

$$\|D\|_2 = \max_{j=0}^{m-1} |\delta_j|.$$

▼ Solution

First, we show that $\|D\|_2 = \max_{\|x\|_2=1} \|Dx\|_2 \leq \max_{i=0}^{m-1} |\delta_i|$:

$$\begin{aligned}
& \|D\|_2^2 \\
&= \quad < \text{definition} > \\
& \max_{\|x\|_2=1} \|Dx\|_2^2 \\
&= \quad < \text{diagonal vector multiplication} > \\
& \max_{\|x\|_2=1} \left\| \begin{pmatrix} \delta_0 \chi_0 \\ \vdots \\ \delta_{m-1} \chi_{m-1} \end{pmatrix} \right\|_2^2 \\
&= \quad < \text{definition} > \\
& \max_{\|x\|_2=1} \sum_{i=0}^{m-1} |\delta_i \chi_i|^2 \\
&= \quad < \text{homogeneity} > \\
& \max_{\|x\|_2=1} \sum_{i=0}^{m-1} |\delta_i|^2 |\chi_i|^2 \\
&\leq \quad < \text{algebra} > \\
& \max_{\|x\|_2=1} \sum_{i=0}^{m-1} \left[\max_{j=0}^{m-1} |\delta_j| \right]^2 |\chi_i|^2 \\
&= \quad < \text{algebra} > \\
& \left[\max_{j=0}^{m-1} |\delta_j| \right]^2 \max_{\|x\|_2=1} \sum_{i=0}^{m-1} |\chi_i|^2 \\
&= \quad < \|x\|_2 = 1 > \\
& \left[\max_{j=0}^{m-1} |\delta_j| \right]^2.
\end{aligned}$$

Next, we show that there is a vector y with $\|y\|_2 = 1$ such that $\|Dy\|_2 = \max_{i=0}^{m-1} |\delta_i|$:

Let j be such that $|\delta_j| = \max_{i=0}^{m-1} |\delta_i|$ and choose $y = e_j$. Then

$$\begin{aligned}
& \|Dy\|_2 \\
&= \quad < y = e_j > \\
& \|De_j\|_2 \\
&= \quad < D = \text{diag}(\delta_0, \dots, \delta_{m-1}) > \\
& \|\delta_j e_j\|_2 \\
&= \quad < \text{homogeneity} > \\
& |\delta_j| \|e_j\|_2 \\
&= \quad < \|e_j\|_2 = 1 > \\
& |\delta_j| \\
&= \quad < \text{choice of } j > \\
& \max_{i=0}^{m-1} |\delta_i|
\end{aligned}$$

Hence $\|D\|_2 = \max_{j=0}^{m-1} |\delta_j|$.

Homework 1.3.5.2. Let $y \in \mathbb{C}^m$ and $x \in \mathbb{C}^n$.

ALWAYS/SOMETIMES/NEVER: $\|yx^H\|_2 = \|y\|_2 \|x\|_2$.

► [Hint](#) ► [Answer](#) ▼ [Solution](#)

W.l.o.g. assume that $x \neq 0$.

We know by the Cauchy-Schwarz inequality that $|x^H z| \leq \|x\|_2 \|z\|_2$. Hence

$$\begin{aligned}
 & \|yx^H\|_2 \\
 &= \text{< definition >} \\
 & \max_{\|z\|_2=1} \|yx^H z\|_2 \\
 &= \text{< } \|\cdot\|_2 \text{ is homogeneous >} \\
 & \max_{\|z\|_2=1} |x^H z| \|y\|_2 \\
 &\leq \text{< Cauchy-Schwarz inequality >} \\
 & \max_{\|z\|_2=1} \|x\|_2 \|z\|_2 \|y\|_2 \\
 &= \text{< } \|z\|_2 = 1 \text{ >} \\
 & \|x\|_2 \|y\|_2.
 \end{aligned}$$

But also

$$\begin{aligned}
 & \|yx^H\|_2 \\
 &= \text{< definition >} \\
 & \max_{z \neq 0} \|yx^H z\|_2 / \|z\|_2 \\
 &\geq \text{< specific } z \text{ >} \\
 & \|yx^H x\|_2 / \|x\|_2 \\
 &= \text{< } x^H x = \|x\|_2^2; \text{ homogeneity >} \\
 & \|x\|_2^2 \|y\|_2 / \|x\|_2 \\
 &= \text{< algebra >} \\
 & \|y\|_2 \|x\|_2.
 \end{aligned}$$

Hence

$$\|yx^H\|_2 = \|y\|_2 \|x\|_2.$$

Homework 1.3.5.3. Let $A \in \mathbb{C}^{m \times n}$ and a_j its column indexed with j .

ALWAYS/SOMETIMES/NEVER: $\|a_j\|_2 \leq \|A\|_2$.

► [Hint](#)

► [Answer](#)

▼ [Solution](#)

$$\begin{aligned}
 & \|a_j\|_2 \\
 &= \\
 & \|Ae_j\|_2 \\
 &\leq \\
 & \max_{\|x\|_2=1} \|Ax\|_2 \\
 &= \\
 & \|A\|_2.
 \end{aligned}$$

Homework 1.3.5.4. Let $A \in \mathbb{C}^{m \times n}$. Prove that

- $\|A\|_2 = \max_{\|x\|_2=\|y\|_2=1} |y^H Ax|$.
- $\|A^H\|_2 = \|A\|_2$.
- $\|A^H A\|_2 = \|A\|_2^2$.

► [Hint](#)

▼ [Solution](#)

- $\|A\|_2 = \max_{\|x\|_2=\|y\|_2=1} |y^H Ax|$:

$$\begin{aligned}
 & \max_{\|x\|_2=\|y\|_2=1} |y^H Ax| \\
 & \leq \quad < \text{Cauchy-Schwarz} > \\
 & \max_{\|x\|_2=\|y\|_2=1} \|y\|_2 \|Ax\|_2 \\
 & = \quad < \|y\|_2 = 1 > \\
 & \max_{\|x\|_2=1} \|Ax\|_2 \\
 & = \quad < \text{definition} > \\
 & \|A\|_2.
 \end{aligned}$$

Also, we know there exists x with $\|x\|_2 = 1$ such that $\|A\|_2 = \|Ax\|_2$.
Let $y = Ax / \|Ax\|_2$. Then

$$\begin{aligned}
 & |y^H Ax| \\
 & = \quad < \text{instantiate} > \\
 & \left| \frac{(Ax)^H (Ax)}{\|Ax\|_2} \right| \\
 & = \quad < z^H z = \|z\|_2^2 > \\
 & \left| \frac{\|Ax\|_2^2}{\|Ax\|_2} \right| \\
 & = \quad < \text{algebra} > \\
 & \|Ax\|_2 \\
 & = \quad < x \text{ was chosen so that } \|Ax\|_2 = \|A\|_2 > \\
 & \|A\|_2
 \end{aligned}$$

Hence the bound is attained. We conclude that

$$\|A\|_2 = \max_{\|x\|_2=\|y\|_2=1} |y^H Ax|.$$

- $\|A^H\|_2 = \|A\|_2$:

$$\begin{aligned}
 & \|A^H\|_2 \\
 & = \quad < \text{first part of homework} > \\
 & \max_{\|x\|_2=\|y\|_2=1} |y^H A^H x| \\
 & = \quad < |\bar{\alpha}| = |\alpha| > \\
 & \max_{\|x\|_2=\|y\|_2=1} |x^H A y| \\
 & = \quad < \text{first part of homework} > \\
 & \|A\|_2.
 \end{aligned}$$

- $\|A^H A\|_2 = \|A\|_2^2$.

$$\begin{aligned}
& \|A^H A\|_2 \\
&= \quad < \text{first part of homework} > \\
& \max_{\|x\|_2=\|y\|_2=1} |y^H A^H A x| \\
&\geq \quad < \text{restricts choices of } y > \\
& \max_{\|x\|_2=1} |x^H A^H A x| \\
&= \quad < z^H z = \|z\|_2^2 > \\
& \max_{\|x\|_2=1} \|Ax\|_2^2 \\
&= \quad < \text{algebra} > \\
& \left(\max_{\|x\|_2=1} \|Ax\|_2 \right)^2 \\
&= \quad < \text{definition} > \\
& \|A\|_2^2.
\end{aligned}$$

So, $\|A^H A\|_2 \geq \|A\|_2^2$.

Now, let's show that $\|A^H A\|_2 \leq \|A\|_2^2$. This would be trivial if we had already discussed the fact that $\|\cdot\|_2$ is a submultiplicative norm (which we will in a future unit). But let's do it from scratch. First, we show that $\|Ax\|_2 \leq \|A\|_2 \|x\|_2$ for all (appropriately sized) matrices A and x :

$$\begin{aligned}
& \|Ax\|_2 \\
&= \quad < \text{norms are homogeneous} > \\
& \|A \frac{x}{\|x\|_2}\|_2 \|x\|_2 \\
&\leq \quad < \text{algebra} > \\
& \max_{\|y\|_2=1} \|Ay\|_2 \|x\|_2 \\
&= \quad < \text{definition of 2-norm} > \\
& \|A\|_2 \|x\|_2.
\end{aligned}$$

With this, we can then show that

$$\begin{aligned}
& \|A^H A\|_2 \\
&= \quad < \text{definition of 2-norm} > \\
& \max_{\|x\|_2=1} \|A^H A x\|_2 \\
&\leq \quad < \|Az\|_2 \leq \|A\|_2 \|z\|_2 > \\
& \max_{\|x\|_2=1} (\|A^H\|_2 \|Ax\|_2) \\
&= \quad < \text{algebra} > \\
& \|A^H\|_2 \max_{\|x\|_2=1} \|Ax\|_2 \\
&= \quad < \text{definition of 2-norm} > \\
& \|A^H\|_2 \|A\|_2 \\
&= \quad < \|A^H\|_2 = \|A\|_2 > \\
& \|A\|_2^2
\end{aligned}$$

Alternatively, as suggested by one of the learners in the course, we can use the Cauchy-Schwarz inequality:

$$\begin{aligned}
& \|A^H A\|_2 \\
& = < \text{part (a) of this homework} > \\
& \max_{\|x\|_2=\|y\|_2=1} |x^H A^H A y| \\
& = < \text{simple manipulation} > \\
& \max_{\|x\|_2=\|y\|_2=1} |(Ax)^H Ay| \\
& \leq < \text{Cauchy-Schwarz inequality} > \\
& \max_{\|x\|_2=\|y\|_2=1} \|Ax\|_2 \|Ay\|_2 \\
& = < \text{algebra} > \\
& \max_{\|x\|_2=1} \|Ax\|_2 \max_{\|y\|_2=1} \|Ay\|_2 \\
& = < \text{definition} > \\
& \|A\|_2 \|A\|_2 \\
& = < \text{algebra} > \\
& \|A\|_2^2
\end{aligned}$$

Homework 1.3.5.5. Partition $A = \left(\begin{array}{c|c|c} A_{0,0} & \cdots & A_{0,N-1} \\ \hline \vdots & & \vdots \\ \hline A_{M-1,0} & \cdots & A_{M-1,N-1} \end{array} \right)$.

ALWAYS/SOMETIMES/NEVER: $\|A_{i,j}\|_2 \leq \|A\|_2$.

► [Hint](#) ▼ [Solution](#)

Choose v_j and w_i such that $\|A_{i,j}\|_2 = |w_i^H A_{i,j} v_j|$. Next, choose v and w such that

$$v = \begin{pmatrix} \frac{0}{\vdots} \\ \frac{0}{v_j} \\ \frac{0}{\vdots} \\ \frac{0}{0} \end{pmatrix}, \quad w = \begin{pmatrix} \frac{0}{\vdots} \\ \frac{0}{w_i} \\ \frac{0}{\vdots} \\ \frac{0}{0} \end{pmatrix}.$$

You can check (using partitioned multiplication and the last homework) that $w^H A v = w_i^H A_{i,j} v_j$. Then, by [Homework 1.3.5.4](#)

$$\|A\|_2$$

$$= \text{< last homework >}$$

$$\max_{\|x\|_2=\|y\|_2=1} |y^H Ax|$$

$$\geq \text{< } w \text{ and } v \text{ are specific vectors >}$$

$$|w^H Av|$$

$$= \text{< partitioned multiplication >}$$

$$|w_i^H A_{i,j} v_j|$$

$$= \text{< how } w_i \text{ and } v_j \text{ were chosen >}$$

$$\|A_{i,j}\|_2.$$